

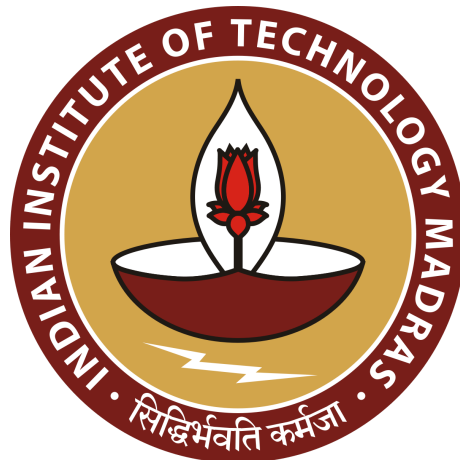
CIVIC SERVICES PLATFORM: COMPLAINT MANAGEMENT SYSTEM

MILESTONE 1

Course: Software Engineering

Team 107

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1. Introduction

Urban civic complaint management continues to face challenges due to fragmented communication channels, limited transparency, manual workflows, and delayed issue resolution. Citizens often struggle to report issues efficiently, while municipal authorities face operational challenges in prioritizing, assigning, and tracking complaints across departments.

This project aims to design and develop a centralized Civic Services Platform that streamlines complaint management for both citizens and municipal authorities. The requirements presented in this document are derived from user identification, interviews, and observation studies conducted with relevant stakeholders.

2. Problem Statement

Many interactions between citizens and local authorities involve inefficient workflows and poor visibility throughout the complaint resolution lifecycle and in some cases proper communication channels are unknown. Existing systems often lack transparency, centralized tracking, timely updates and accountability, resulting in reduced public trust and operational inefficiencies.

The objective of this project is to develop a Complaint Management System that addresses these challenges through improved workflow management, complaint tracking, and communication between citizens and authorities.

3. Objectives

The primary objectives of this study are:

- Identify all stakeholders interacting with the system.
- Understand existing complaint management workflows.
- Identify challenges faced by different stakeholders.
- Document recurring pain points.
- Derive system requirements based on real user needs.
- Formulate SMART user stories for future development.

4. Identification of Users

4.1 User Classification

The stakeholders interacting with the system are classified into three categories based on their level of involvement.

Primary Users

Primary users interact directly with the system on a regular basis.

User	Description
Citizen	Registers complaints, tracks status, receives updates, and provides feedback.
Complaint Desk Operator	Registers complaints on behalf of offline citizens and assists with status inquiries.
Department Officer	Reviews, prioritizes, and manages complaints assigned to their department.
Field Supervisor	Assigns complaints to field workers and monitors execution.
Field Worker	Visits complaint locations, performs work, uploads completion evidence, and updates status.

Secondary Users

Secondary users support operational activities and administrative workflows.

User	Description
Ward/Zone Officer	Monitors complaints within a geographical region and escalates delayed cases.
Grievance Officer	Handles escalated or disputed complaints.
Quality Inspector	Verifies whether reported issues have been resolved satisfactorily.
Vendor/External Contractor	Executes outsourced civic work and submits completion reports.

Tertiary Users

Tertiary users primarily consume reports and monitor overall system performance.

User	Description
Municipal Commissioner / Chairman	Reviews department performance and city-wide complaint trends.
Auditor / Review Committee	Conducts compliance reviews and verifies accountability through audit logs.
System Administrator	Manages users, departments, permissions, and system configuration.

5. User Research Methodology

A qualitative user research approach was adopted to understand the existing complaint management process from both citizen and authority perspectives.

Research Method

- Semi-structured interviews
- Contextual observation
- Qualitative analysis

Participants

Stakeholder	Participants
Citizens	4
Government Authorities	2

Research Objective

The interviews focused on identifying existing challenges, inefficiencies, and user expectations. Participants were encouraged to describe their experiences rather than comment on potential system features.

6. Interview Findings

Summarize the recurring observations under separate headings.

Findings from Citizens

- Lack of visibility into complaint progress.
- No acknowledgement after complaint submission.
- Uncertainty regarding responsible departments.
- Repeated reporting of unresolved issues.
- No estimated resolution timeline.
- Lack of proper communication channels.
- Difficulty reporting complaints due to language barriers.
- Lack of trust in the complaint resolution process.

Findings from Authorities

- Complaints received through multiple disconnected channels.
- Poor quality and incomplete complaint information.
- Manual complaint assignment.
- Duplicate complaint submissions.
- Limited visibility into officer workloads.
- No structured escalation mechanism.
- Difficulty tracking complaint ownership.

7. Observation Study

In addition to interviews, observations of existing complaint reporting workflows revealed several operational inefficiencies.

Key observations include:

- Complaint information is scattered across multiple communication channels.
- Status updates are rarely communicated proactively.
- Citizens frequently rely on repeated follow-ups.
- Officers manually coordinate complaint assignments.
- There is limited visibility into pending or overdue complaints.

9. Pain Point Analysis

Provide a consolidated table summarizing the identified issues.

Pain Point	Affected Users	Business Impact
Lack of complaint tracking	Citizens	Reduced transparency
Manual complaint assignment	Authorities	Increased resolution time
Duplicate complaints	Citizens & Authorities	Operational inefficiency
Missing complaint information	Authorities	Delayed processing
Lack of escalation workflow	Authorities	Complaints remain unresolved
No accountability	Citizens	Reduced public trust

10. Functional Requirements Derived from User Research

The identified pain points were translated into high-level functional requirements.

Examples include:

- Complaint submission with supporting evidence.
- Complaint status tracking.
- Automatic department assignment.
- Officer workload management.
- Complaint escalation workflow.
- Duplicate complaint detection.
- Citizen notifications.
- Resolution feedback mechanism.
- Administrative dashboards.
- Audit logging.

11. User Stories

The following user stories have been derived from the user research interviews and observation studies. Each story follows the SMART (Specific, Measurable, Achievable, Relevant, and Time-bound) principle and captures a clear user objective along with the value it provides.

Citizen

User Story ID	User Story
US-01	As a citizen, I want to submit a complaint by providing its category, location, description, and supporting images, so that the concerned department has sufficient information to begin resolving the issue promptly.
US-02	As a citizen, I want to track the status of my complaint throughout its lifecycle, so that I remain informed about its progress without having to contact the authorities.
US-03	As a citizen, I want to receive notifications whenever the status of my complaint changes, so that I stay updated on actions taken by the authorities.
US-04	As a citizen, I want to provide feedback after my complaint is resolved, so that I can indicate whether the issue has been addressed satisfactorily.

Complaint Desk Operator

User Story ID	User Story
US-05	As a Complaint Desk Operator, I want to register complaints on behalf of citizens who visit or call the office, so that offline users can also access civic services.
US-06	As a Complaint Desk Operator, I want to search complaints using the complaint ID, citizen name, or phone number, so that I can quickly respond to citizen inquiries.
US-07	As a Complaint Desk Operator, I want to update incomplete complaint information whenever additional details are received, so that complaints can be processed without delays.

Department Officer

User Story ID	User Story
US-08	As a Department Officer, I want to review all complaints assigned to my department, so that I can prioritize and manage them efficiently.
US-09	As a Department Officer, I want to assign complaints to the appropriate field supervisor, so that work can begin without unnecessary delays.
US-10	As a Department Officer, I want to approve or reject complaint closures based on submitted evidence, so that complaints are closed only after satisfactory resolution.

Field Supervisor

User Story ID	User Story
US-11	As a Field Supervisor, I want to assign complaints to available field workers based on workload and expertise, so that work is distributed efficiently.
US-12	As a Field Supervisor, I want to monitor the progress of complaints assigned to my team, so that delays can be identified and addressed promptly.
US-13	As a Field Supervisor, I want to reassign unresolved complaints when required, so that service delivery remains uninterrupted.

Field Worker

User Story ID	User Story
US-14	As a Field Worker, I want to view all complaints assigned to me along with their location and description, so that I can reach the site prepared for the required work.
US-15	As a Field Worker, I want to update the complaint status while performing the assigned work, so that supervisors and citizens receive accurate progress updates.
US-16	As a Field Worker, I want to upload photographs after completing the work, so that there is verifiable evidence of complaint resolution.

Ward / Zone Officer

User Story ID	User Story
US-17	As a Ward / Zone Officer, I want to monitor all complaints within my jurisdiction, so that I can identify recurring civic issues and ensure timely resolution.
US-18	As a Ward / Zone Officer, I want to identify complaints that have exceeded their resolution deadlines, so that appropriate escalation measures can be initiated.

Grievance Officer

User Story ID	User Story
US-19	As a Grievance Officer, I want to review complaints that have been escalated or reopened, so that unresolved citizen concerns receive additional attention.
US-20	As a Grievance Officer, I want to reopen complaints when citizens are dissatisfied with the resolution, so that unresolved issues can be addressed appropriately.

Quality Inspector

User Story ID	User Story
US-21	As a Quality Inspector, I want to verify the evidence submitted after complaint resolution, so that I can ensure the work has been completed according to the required standards.
US-22	As a Quality Inspector, I want to reject incomplete or poor-quality work, so that complaints are not closed until the issue has been resolved satisfactorily.

Vendor / External Contractor

User Story ID	User Story
US-23	As a Vendor / External Contractor, I want to receive work orders assigned by the department, so that outsourced civic maintenance tasks can be completed efficiently.
US-24	As a Vendor / External Contractor, I want to submit work completion reports and supporting evidence, so that the department can verify the completed work.

Municipal Commissioner / Chairman

<i>User Story ID</i>	<i>User Story</i>
US-25	As the Municipal Commissioner / Chairman, I want to view city-wide complaint analytics and departmental performance reports, so that I can make informed administrative decisions.
US-26	As the Municipal Commissioner / Chairman, I want to monitor unresolved and overdue complaints across departments, so that accountability and service quality are maintained.

Auditor / Review Committee

<i>User Story ID</i>	<i>User Story</i>
US-27	As an Auditor / Review Committee Member, I want to review complaint histories and system audit logs, so that I can verify transparency, compliance, and accountability within the complaint management process.
US-28	As an Auditor / Review Committee Member, I want to generate audit reports summarizing complaint resolution activities, so that governance and operational performance can be evaluated effectively.

System Administrator

User Story ID	User Story
US-29	As a System Administrator, I want to create and manage user accounts, roles, and permissions, so that only authorized users can access the system.
US-30	As a System Administrator, I want to configure departments, complaint categories, and routing rules, so that complaints are directed to the appropriate authorities.
US-31	As a System Administrator, I want to monitor system activity and maintain audit logs, so that operational integrity, security, and accountability can be ensured.
US-32	As a System Administrator, I want to configure notification templates and system settings, so that users receive timely and consistent communication throughout the complaint lifecycle.

12. Conclusion

The user research successfully identified the major operational challenges faced by citizens and municipal authorities in the current complaint management process. Rather than assuming software features, the study focused on understanding real-world workflows, recurring frustrations, and operational bottlenecks.

The insights gathered through interviews and observation directly informed the functional requirements and user stories presented in this report. These findings will serve as the foundation for the subsequent design, implementation, and testing phases of the Civic Services Platform.